

Basics

decay time of μ is smallest in its own rest frame: $\tau_{lab} = \gamma \tau_{rest}$

proper velocity: $\eta_x = dx/d\tau_{rest}$, $u = \frac{dx}{dt} = \frac{dx'}{dt'} \frac{dt'}{dt} = \frac{u' + v}{1 + \frac{v u'}{c^2}}$ $\hookrightarrow L = v \tau_{lab} = \beta \gamma c \tau$

trick: square both sides when solving for smt.

total energy: $E^2 = p^2 c^2 + m^2 c^4 = p^2 + m^2$ (N.U.)

alg.: $\lambda = \frac{hc}{E}$

$\tan \theta_{lab} = \frac{p_{lab}}{E_{lab}} = \frac{p \sin \theta}{E \sin \theta} = \frac{p \sin \theta}{E \sin \theta}$

Relativistic kinematics:

norm of 4-momentum is Lorentz-inv.: $P_\mu P^\mu = \left(\frac{E/c}{\vec{p}} \right) \left(\frac{E/c}{\vec{p}} \right) = \frac{E^2}{c^2} - \vec{p}^2 = m^2 c^2$

Invariant mass: $\frac{E_{cm}}{c} = \sqrt{s} = \sqrt{P_\mu P^\mu} = \sqrt{(E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2}$. Easier to calculate P_{cm} & boost: $P_{lab} = \Lambda P_{cm}$

(CMS: total momentum vanishes, single photon can't be boosted.

$E_{kin} \ll m$: classical approx. $p = \sqrt{2mE_{kin}}$. $E_{kin} \gg m$: ultra-rel. app. $p = E_{kin}$

Collider vs. fixed target: to have same E_{cm} , we need a higher beam energy in fixed target than in collider.

produced particles have large E_{kin} : $P_1 = (E_1, \vec{p}_1)$, $P_2 = (E_2, \vec{p}_2) \rightarrow P = (E_1 + E_2, \vec{p}_1 + \vec{p}_2) \rightarrow \sqrt{s} = \sqrt{m_1^2 + m_2^2 + 2E_1 E_2}$

0.9c $\rightarrow$ SI $\vec{v}$: $P_1 = (E_1, \vec{p}_1)$, $P_2 = (E_2, \vec{p}_2) \rightarrow \sqrt{s} = \sqrt{m_1^2 + m_2^2 + 2E_1 E_2 - 2\vec{p}_1 \cdot \vec{p}_2}$

relativ.: $E = \sqrt{m^2 c^4 + p^2 c^2} = \gamma m c^2$, $p = \gamma m v$, $E_{kin} = E - E_0$; non-relativ.: $E = m c^2 + \frac{p^2}{2m}$, $p = m v$

$v = \frac{p}{m \gamma} \rightarrow \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$ $E_{kin}^{min}: E_{cm} = \text{outgoing masses}$

Lorentz Force: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \rightarrow \vec{B} \text{ const.} \& \vec{v} \perp \vec{B}: p = qRB \times R [m] = 3.3 \frac{p [GeV]}{q [e] B [T]}$

$\beta = N(t) - N_0 = \# \text{ of } e^-$ decay rate: $N(t) = N_0 e^{-\lambda t}$. Mean lifetime $\tau = \frac{1}{\lambda}$, Half life time $t_{1/2} = \frac{\ln(2)}{\lambda}$

Activity: # of decays/s: $A(t) = \lambda N(t) = \lambda N_0 e^{-\lambda t} \rightarrow A = -\frac{dN}{dt} = \lambda N$ Uncertainty: $\Delta E \Delta t > \frac{\hbar}{2} \rightarrow \Delta E \sim \frac{\hbar}{2 \tau}$

Allowed? For energy, check masses $N_0 = \Delta t \cdot (\phi(0) - \phi(x))$ mean free path: $\lambda = \text{Length} / N_{col}$ $N_{col} = \# \text{ of coll.} = \frac{\Delta E}{E}$

Decay rate: $\frac{dN}{dt} = \text{Power}$ and N (# of decayed nuclei): $N = \frac{M \cdot N_A}{A \cdot n} \rightarrow E_{ex}: ^{239}\text{Pu} \rightarrow ^{235}\text{U} + \alpha$

Recoil energy: $E = \frac{p_\alpha^2}{2M_\alpha} = \frac{p_\alpha^2}{2M_\alpha} = \frac{2M_\alpha E_\alpha}{2M_\alpha} \rightarrow E_{kin}$ released per α -decay: $E = E_\alpha + E_\alpha = \frac{A_{start}}{A_{end}} \cdot E_\alpha$

$\hookrightarrow N_0 = \frac{m}{N} \cdot N_A \rightarrow \text{emitted particles: } N_e = N_0 - N(t)$ 2-step-decay: $\frac{dN_B}{dt} = -\lambda_A N_A(t) + \lambda_B N_B(t)$

Natural Units: $1 \text{ kg} = 5.61 \cdot 10^{26} \text{ GeV}$, $\hbar c = 197.327 \text{ MeV fm} \rightarrow 1 \text{ m} = 5.07 \cdot 10^{15} (\text{GeV})^{-1}$, $1 \text{ s} = 1.52 \cdot 10^{24} (\text{GeV})^{-1}$

Lab Frame Center of momentum (CM) frame

Dating: $\Delta t = t_1 - t_0 = \frac{1}{\lambda} \ln(1 + \frac{N_0(t_1)}{N_0(t_2)})$ $\times$ $N_{parent}(t_1) = N_0(t_0) e^{-\lambda(t_1 - t_0)}$, $N_p(t_1) = N_0(t_1) + N_d(t_1)$

$\vec{p}_a + \vec{p}_b = 0$ $\vec{p}_a + \vec{p}_b = 0$ $\vec{p}_a + \vec{p}_b = 0$ $\vec{p}_a + \vec{p}_b = 0$

Atomic Physics ($> 180^\circ \rightarrow 0^\circ - 360^\circ$) $\star$ energy gain from fission/fusion $\sim \text{MeV} \gg \text{oil/gas} (\sim \text{eV}) \rightarrow \text{more efficient}$

Some bound neutrons don't decay due to energy imbalance It's not easy for p to penetrate nucleus ($> 1 \text{ fm} \rightarrow \text{EM dominates, Coulomb barrier}$)

Fusion products have too few n to produce a stable nucleus & to convert $p \rightarrow n + e^+ + \nu_e$. There is no bound state of $2p$. $^1_1\text{H} + ^1_1\text{H} \rightarrow ^2_2\text{He} + Q$ pp -fue. need $E = \frac{e^2}{4\pi\epsilon_0 R}$ $\times$ $E = \frac{1}{2} kT \rightarrow$ but lower T happens too: only av. E per proton. MB-distr.

has a tail $\rightarrow$ some p have very large E . Quantum tunneling: p don't need to overcome Coulomb barrier entirely.

Fission products have more N than their stable isobars $\rightarrow$ unstable against β -decay. Branching ratio: $\Gamma(^{241}\text{Pu} \rightarrow ^{241}\text{Am} + \dots) \propto (E_{\beta})^5$

Mössbauer Effect: Recoil energy negligible. Accurate method for measuring the widths of nuclear transitions. Fix source & target to a crystal.

$\hookrightarrow$ excitation energy $E_0 = E_\gamma + \frac{E_\gamma^2}{2M_N c^2}$ Doppler shift: $\Gamma = 2E \frac{v_{1/2}}{c}$

Protons bind less tightly than neutrons since they have to overcome Coulomb repulsion. $\rightarrow N \neq P$ bc. of asymmetry up to $A \sim 20$.

Liquid Drop Model: predicts binding energies for: $N = (A - Z)$, $Z = 2, 8, 20, \dots$ $\rightarrow$ neutron sep. energy. If N is a MN, then σ for n -absorp.

is much lower than for other nuclides. E of excited states $\gg E_0$ if $N \neq$ or Z are MN.

$Z = MN$: larger, natural abundance $\star$ can't explain asymmetry term ($Z \neq N$: small $\rightarrow$ Fermi gas theory, less important for large A)

mass $\rightarrow$ 2.3 MeV/c² $\rightarrow$ 1.275 GeV/c² $\rightarrow$ 1.737 GeV/c² $\rightarrow$ 0 $\rightarrow$ 126 GeV/c²

charge $\rightarrow$ 2/3 $\rightarrow$ 2/3 $\rightarrow$ 2/3 $\rightarrow$ 0 $\rightarrow$ 0

spin $\rightarrow$ 1/2 $\rightarrow$ 1/2 $\rightarrow$ 1/2 $\rightarrow$ 0 $\rightarrow$ 0

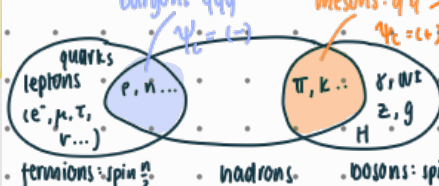
up $\rightarrow$ charm $\rightarrow$ top $\rightarrow$ gluon $\rightarrow$ Higgs boson

down $\rightarrow$ strange $\rightarrow$ bottom $\rightarrow$ photon

electron $\rightarrow$ muon $\rightarrow$ tau $\rightarrow$ Z boson

electron neutrino $\rightarrow$ muon neutrino $\rightarrow$ tau neutrino $\rightarrow$ W boson

CP symm. slightly broken (12)



The coupling strength between Higgs & a fermion is determined by the fermions mass

Higgs doesn't couple to g, γ & ν , t quark coupled strongest to H (Yukawa)

anti-part. same m & τ don't interact by $e-w$. $\rightarrow$ no mass

SM always conserves: Baryon & Lepton no., angular mom., el. charge & CPT

Interaction Rates

Flux of particle: $\phi_a = \frac{N_a}{\Delta t A} = \frac{\# \text{ of part.}}{\Delta t \cdot \text{Area of interest}} = n_a v_a$ (particle density)
 Reaction rate: $N = \frac{\Delta N}{\Delta t} = \phi_a N_b \sigma = L \sigma_{tot}$
 Luminosity: $L = \phi_a N_b = n_a v_a N_b = \phi_a \int dV n_b = \phi_a \int dV \frac{N_b}{V} = \phi_a \frac{N_b}{V} \int dV = \phi_a N_b$
 The increase of activated nuclei during activation process is given by: $\frac{dN}{dt} = \sigma \phi_a N_b A = \sigma \phi_a N_b \frac{A}{M} \times A \rightarrow N_a N_b \sigma \int dV / (A M)$

Total cross section: Measures probability of a collision as an effective area. $\sigma = \frac{N}{N_b \phi_a} = \frac{N}{L}$
 Diff. cross section: $\frac{d\sigma}{d\Omega dE'}$: we can look at specific regions. $N = L \frac{d\sigma}{d\Omega dE'} \Delta\Omega \Delta E'$
 Fermi's Golden Rule: Expresses a physically measurable quantity (dσ) to the physics described by ψ_i & the density of the phase space, with kinematic constraints.

$\psi_i = \frac{1}{\sqrt{V}} e^{i\mathbf{p}_i \cdot \mathbf{r}}$
 reaction rate, width ... $\Gamma = \frac{1}{\tau} = \frac{1}{N} \frac{dN}{dt}$
 $\Gamma = \frac{2\pi}{\hbar} |\langle \psi_f | H' | \psi_i \rangle|^2 \rho(E_f)$
 reaction rate: $\frac{d\sigma}{d\Omega dE}$
 $\rho(E) = \frac{1}{(2\pi)^3} \frac{d^3p}{dE}$
 How many configurations of momentum are in the final state?
 Sargent's Rule: $\rho(E) \propto E^5$
 2-body $\pi \rightarrow e^+ e^-$ decay: $\Gamma(E_0) \propto \frac{(m_\pi^2 - m_e^2)(m_\pi^2 - m_e^2)^2}{m_\pi^2}$
 m_π : Physical (on-shell) mass of exchanged particle X.
 invariant!

F.T. of potential $\rightarrow$ propagator $\frac{1}{q^2 - m^2}$
 $q^2 = (E_1 - E_2)^2 - (\mathbf{p}_1 - \mathbf{p}_2)^2 = (E_1 - E_2)^2 - (\mathbf{p}_1^2 - \mathbf{p}_2^2) = (E_1 - E_2)^2 - (\mathbf{p}_1^2 - \mathbf{p}_2^2)$
 at low E $\approx$ const. (Fermi-C.)
 $\psi_i = \frac{1}{\sqrt{V}} e^{i\mathbf{p}_i \cdot \mathbf{r}}$
 $\psi_f = \frac{1}{\sqrt{V}} e^{i\mathbf{p}_f \cdot \mathbf{r}}$
 $\langle \psi_f | H' | \psi_i \rangle = \frac{1}{V} \int d^3r e^{i(\mathbf{p}_f - \mathbf{p}_i) \cdot \mathbf{r}} V(r)$
 $\rightarrow \delta(\mathbf{p}_f - \mathbf{p}_i)$
 $\rightarrow$ momentum exchange (off-shell mass)

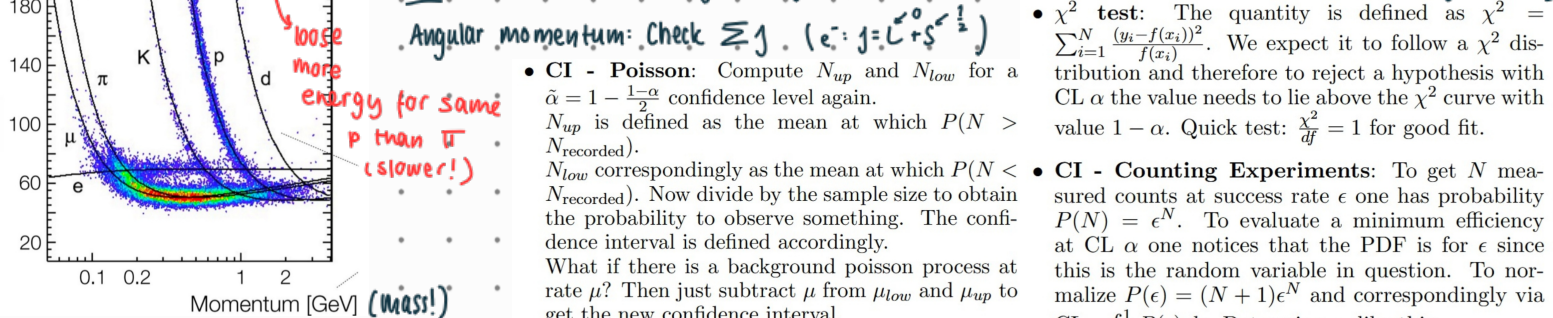
Low E decay: $q^2_{max} < M_W^2 \rightarrow \Gamma_{H \rightarrow W} \propto G_F^2 (M_H^2 - M_W^2)^2$
 Number of detected events: $M_{meas} = L \sigma_{prod} \epsilon$
 efficiency $\epsilon = \int_0^1 d\epsilon$
 Z-Boson: $e^+ e^- \rightarrow Z$
 Pair production cross section at $\sqrt{s} < 2m$ than at nuclei with charge Z: $\sigma_{pp} \sim \frac{1}{4} \sigma_{Z} \sim \frac{1}{4} \frac{Z^2}{s}$
 ipace & propagators are the same. Atom composed of nucl. with charge Ze & Z e-
 $\rightarrow$ incoherent sum: $\frac{\text{prob. of pair prod. at nuclei}}{\text{prob. of pp at atomic e-}} = \frac{\sigma_{pp}}{\sigma_{Z}} = Z^2$

Partial decay width: $\tau = \frac{\hbar}{\Gamma}$
 $\Gamma_{tot} = \Gamma_e + \Gamma_\mu + \Gamma_\tau + \Gamma_\gamma + \Gamma_{had}$
 Experimental difficulties when det. m_ν : the β -spectrum falls @ endpoint $\rightarrow$ low counting rate, big error; finite detector resolution; internal e^- scattering $\rightarrow$ more different E_e . Decay product: excited state with diff. Binding-E $\rightarrow$ superposition
 Rutherford experiment: $\frac{d\sigma}{d\Omega}$ falls with scattering angle, but at large θ it's larger than expected
 Mass spectrometry: $\frac{m}{Q} = \frac{(R_B B)^2}{r_e E}$
 $\rightarrow B = M_H + M_{H-} - M_{H-} = E_\gamma + \frac{E_\gamma^2}{2M_H}$
 negligible for small E_γ & heavy nuclei (e^- scattering)

Statistics
 Mean: $\mu = \int x p(x) dx$, std. $\sigma = \sqrt{\int (x-\mu)^2 p(x) dx}$, FWHM = $2\sqrt{2\ln 2} \sigma$ (Gaussian)
 Binomial distr.: $p(k \text{ out of } n) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$
 Mean # of obs. decays: $\mu = k p$, $\sigma = \sqrt{k p (1-p)}$
 Poisson distr.: Mean # of observed decays: $\mu = \lambda$ (const. for $t \rightarrow \infty$). $P(k, \lambda) = \frac{\mu^k}{k!} e^{-\mu}$
 measurement: $\sigma_x = \frac{\sigma}{\sqrt{n}}$
 Discovery significance: Counting experiment μ_{signal} & $\mu_{background}$ (Poisson) $\rightarrow$ Gaussian: $Z \sqrt{\mu_B}$: discovery significance of Z_0 : $\mu_s \geq Z_0 \sqrt{\mu_B}$
 CI - Base: N groups of data, confidence level α . In N groups, to 95% CI contains μ .

Ex. $A+B \rightarrow C+D$. Parity: multiply (± 1) on both sides: Check Multiplets. Energy: masses, Strong Isospin: I_3
 Angular momentum: Check $\sum J$ ($e^-: j = \ell + s = \frac{1}{2}$)
 CI - Poisson: Compute N_{up} and N_{low} for a $\hat{\alpha} = 1 - \frac{1-\alpha}{2}$ confidence level again.
 N_{up} is defined as the mean at which $P(N > N_{recorded})$.
 N_{low} correspondingly as the mean at which $P(N < N_{recorded})$. Now divide by the sample size to obtain the probability to observe something. The confidence interval is defined accordingly.
 What if there is a background poisson process at rate μ ? Then just subtract μ from μ_{low} and μ_{up} to get the new confidence interval.

χ^2 test: The quantity is defined as $\chi^2 = \sum_{i=1}^N \frac{(y_i - f(x_i))^2}{f(x_i)}$. We expect it to follow a χ^2 distribution and therefore to reject a hypothesis with CL α the value needs to lie above the χ^2 curve with value $1 - \alpha$. Quick test: $\frac{\chi^2}{df} = 1$ for good fit.
 CI - Counting Experiments: To get N measured counts at success rate ϵ one has probability $P(N) = \epsilon^N$. To evaluate a minimum efficiency at CL α one notices that the PDF is for ϵ since this is the random variable in question. To normalize $P(\epsilon) = (N+1)\epsilon^N$ and correspondingly via $CL = \int_0^1 P(\epsilon) d\epsilon$. Determine ϵ_0 like this.



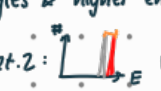
Scattering experiments

★ Mass difference. ($\pi^0, n/p$): cancellation of degeneracy by EM interaction ★

Elastic scattering: 

with recoil: recoil energy of nucleus: $(E - E')$. the larger $\frac{E_0}{M}$, the more recoil Energy is transferred to the target. important at larger angles & higher energies.

without recoil: E_{kin} of projectile is conserved. $(\frac{d\sigma}{d\Omega})(q^2)$ looks the same.

det. 1:  det. 2:  E_{kin} of projectile depends on recoil loss (θ).

$\frac{d\sigma}{d\Omega} \propto \frac{1}{q^4} \propto \frac{1}{\sin^4(\theta/2)}$ (phase space)

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Ex.: If we want to study the electric charge distr. of Ca^{40} -nucleus $\rightarrow$ elastic e^- scattering ($\%: Ru$; nuclear eff. & inel.; smt changes w/ e^- nuclei.)

Scattering on extended charge distribution: Form factor $F(q) = \int \rho(r) e^{-iq \cdot r} d^3r$. Scattering on π -charges creates same diffraction pattern.

FT of charge function: $g(r) = Ze f(r)$ static distr. $\int f(r) d^3r = 1$

limited range of q^2 : Beam energy & q^2

Strangeness found by $p\bar{p}$ collisions $\rightarrow$ hadronic particles: in pairs, large lifetime ($\sim 10^{-8}$ instead of 10^{-24} s).

Low-energy elastic scattering of e^-

Rutherford cross section: $(\frac{d\sigma}{d\Omega})_{Ruth.} = \frac{Z^2 \alpha^2 \hbar^2}{4E^2 \sin^4(\theta/2)}$

Mott cross section: Spin dependence. $(\frac{d\sigma}{d\Omega})_{Mott} = (\frac{d\sigma}{d\Omega})_{Ruth.} (1 - \beta^2 \sin^2(\theta/2))$

ignores finite size of nucleons, is based on relativ. kin.

Backscattering: Spin has to flip. Conserved quantity: $\hbar = \frac{L \cdot p}{|h|p|}$ helicity

depends on nucleus

Deviation: 

Exp. measured: $(\frac{d\sigma}{d\Omega})_{exp} = (\frac{d\sigma}{d\Omega})_{Mott} \cdot |F(q^2)|^2$

Nuclear shapes: oscillating pattern when meas. large, spherical $\rightarrow$ Fermi-distr.: diffuse r. $g(r)$ nonhomog.

RMS charge radius: $R_N = \sqrt{\langle r^2 \rangle} = \sqrt{\int r^2 f(r) d^3x}$

With recoil: $(\frac{d\sigma}{d\Omega})_{Mott} = (\frac{d\sigma}{d\Omega})_{Ruth.} \cdot \frac{E'}{E} \rightarrow q^2 \approx -\frac{4EE'}{c^2} \sin^2(\theta/2)$ (not further simplifiable)

charge distr. $g(r)$ decreases if A incr.

can be eliminated if atom is fixed to a macroscopic body (crystal lattice) $\rightarrow$ Mössbauer effect

high q^2 (larger θ) $\rightarrow$ λ decreases $\rightarrow$ better resolution: $\lambda = \frac{h}{p} = \frac{h}{2M \sin(\theta/2)}$

proof that $u \neq p$ are not points $\rightarrow$ $\frac{g_E(q^2)}{2M^2} = \frac{g_M(q^2)}{2M^2} \tan^2(\theta/2)$

Elastic scattering on single nucleus: Magnetic moment correction: $(\frac{d\sigma}{d\Omega})_{Mott} = (\frac{d\sigma}{d\Omega})_{Ruth.} [1 + \frac{2}{3} \frac{q^2}{M^2} \tan^2(\theta/2)]$

even more precise: Rosenbluth formula: $(\frac{d\sigma}{d\Omega})_{Rb} = (\frac{d\sigma}{d\Omega})_{Mott} \cdot [\frac{G_E^2(q^2) + \tau G_M^2(q^2)}{1 + \tau} + \tau G_M^2(q^2) \tan^2(\theta/2)]$

(based on relativ. kinematics, accounts for recoil & finite size of nucleons)

elastic peak: Incoming E_0 (at θ) $(\frac{d\sigma}{d\Omega})_{el.} = \frac{1}{4} \frac{E_0}{E'} = \frac{1}{4} \frac{E_0}{E_0 - E_0 \sin^2(\theta/2)}$

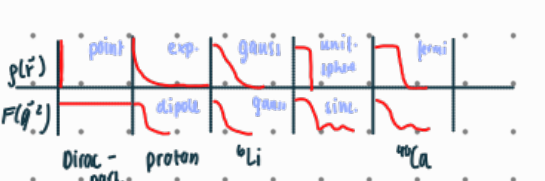
to find true structure of proton: higher energies, inelastic regime

In e^+p scattering, no Δ^{++} resonance $\rightarrow$ charge conservation

Δ^+ (wie proton) scattering process mediated by γ^* $\rightarrow$ no charge exchanged

Initial e^- energy must be > 4.5 GeV. produced e^- : $E_{kin} \sim 0.3$ GeV smaller than elastically scattered e^-

4.5 GeV @ angle where we measure



measured cross sections $\rightarrow$ Form factor

$F(q^2) \xleftrightarrow{F.T.} f(r) = \frac{1}{(2\pi)^3} \int F(q^2) e^{-iq \cdot r} d^3q$, radial distribution

Neutrinos: Flavour ES, not mass ES.

$M_{\nu\mu}$ can be obtained by meas. μ -momentum from π^+ decay @ rest & knowing M_{π^+}, M_{μ^+}

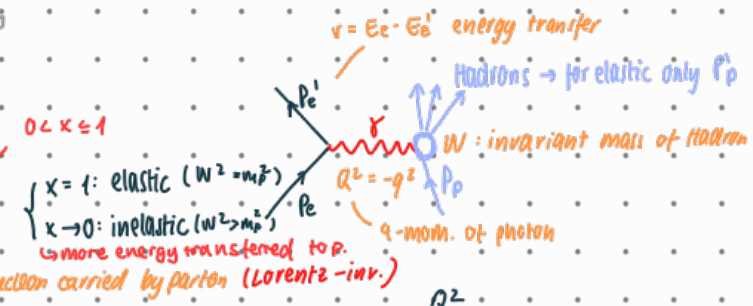
Neutrino oscillations: can occur when $m_{\nu} \neq 0$, they change flavour. Discovery via inverse β -decay: $\bar{\nu}_e + p \rightarrow n + e^+$ (low σ , rare), $e^+ \rightarrow 2\gamma$

$\nu \leftrightarrow \bar{\nu}$ can't be distinguished Cosmic rays (p , high E) $\rightarrow \pi$ (via $\alpha_{str.}$) $\rightarrow$ atmospheric ν . Get m_{ν} by β decay of ^3_1He

Inelastic regime: don't know how much energy e^- will lose approx: proton most likely to break up as $x \rightarrow 0$ & $q \rightarrow 0$

Hadronic resonances: excited states of nucleons which decay via strong interactions. Resonances occur at specif. momentum transfer (i.e. invariant mass)

Bjorken scale & structure functions: $x = \frac{Q^2}{2P \cdot q} = \frac{Q^2}{Q^2 + W^2 - m_p^2}$
analogue to form factors



for small x : gluons are dominant, valence quarks less. Target at rest: $x = \frac{Q^2}{2m_p \nu}$ ($\nu = E_e - E_e'$)
the prob. that a quark is ejected from a proton (e^- -p scattering) increases as $x \rightarrow 0$

at high momentum ($Q^2 \sim 10 \text{ (GeV/c)}^2$)

Bjorken scaling: pointlike objects in proton: (indep. on $Q^2 \rightarrow$ deep inelastic)

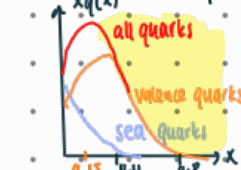
$F_2(x, Q^2)$ max.: hitting a parton with momentum fraction $x \approx 0.2$ most likely
 $\rightarrow$ structure funct. depends only weakly on Q^2 $\leftrightarrow$ structure functions are $\sim$ const. vs. Q^2
 $\rightarrow$ regime: e^- scatter on int. of point-like inside proton
 $\rightarrow Q = \text{const.} \rightarrow$ scaling behavior

Callan-Gross relation: Investigating contribution of p to structure funct. $\rightarrow$ spin of constituents ($\frac{1}{2}$) (need 2 indep. variables to define kinematics for given E_0)
We measure $2x F_1(x) = F_2(x)$ of $2x^2$ & momentum fraction at x . (momentum distr.) $\rightarrow \delta$ -funct.

Proton structure: Elastic form factors, Resonances, deep inelastic, Bjorken & Callan-Gross. $\rightarrow$ Parton model:

elastic: only θ
in fast systems: transv. $\vec{p} \times$ rest mass $= 0$
Elastic interaction with partons
 $Q^2 \gg m^2$
 e^- -p interaction: Σ indep. interact. w/ partons
 q carry more momentum than $\bar{q}$ (valence)

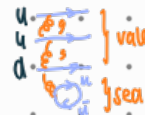
$\langle x_{val} \rangle \sim 0.15$, $\langle x_{sea} \rangle \sim 0.04$ (pp collision: q carry only 15% of momentum)
pp: all q at val! $p: u, d, val$, $\bar{p}: \bar{u}, \bar{d}, val$.



(momentum fraction of proton)
 $f_q = \int_0^1 x q(x) dx \rightarrow f = \int_0^1 dx (x \bar{q}(x) + x q(x))$
 $\hookrightarrow$ predict scattering σ for other processes

fraction carried by gluons: $1 - f \sim 50\%$
low E
pp $\rightarrow$ z^0 : $E_{p, min}$ occurs if q & $\bar{q}$ carry max. possible momentum fraction
 $\rightarrow$ reverse in $\bar{p} \rightarrow$ lower E -thresh.
pp: (occur by colliding val of p & val of $\bar{q}$)
elastic scattering

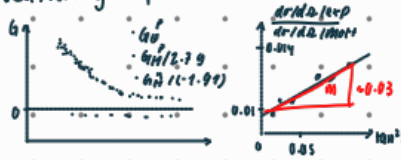
$e^- p \rightarrow e^- X$: elastic: p is recoiling as a single object: $W^2 = M^2$
inelastic: $W^2 = (P + q)^2 = P^2 + 2P \cdot q + q^2 \leftrightarrow M^2 = M^2 - Q^2 + 2M\nu \leftrightarrow Q^2 = 2M\nu$



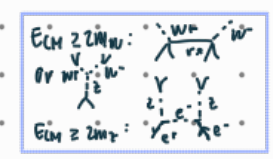
Min. beam energy:

pp: $E_p \approx \frac{m_W^2}{2 \langle x_{val} \rangle} \sim 270 \text{ GeV}$ ($m_W \ll E_p, P_{e^+} \sim 0$, head-on coll.)
pp: $E_p \approx \frac{m_W^2}{2 \sqrt{\langle x_{val} \rangle \langle x_{sea} \rangle}} \sim 570 \text{ GeV}$
Rosenbluth: $q = m_W + q$
 $2 \tau \frac{d^2 \sigma}{d^2 \Omega dE} \sim (6 \frac{1}{2} (Q^2) + C \frac{1}{2} (Q^2) / (1 + \epsilon))$
 $\hat{s} = (P_q + P_{\bar{q}})^2 \sim 4 E_q E_{\bar{q}} \sim q^2 (\langle x_{val} \rangle)^2 E_p^2$
Sjöström: $G_F^2 \sim 0 \rightarrow e^-$ on n , $G_F^2(Q^2) = \frac{m}{2\tau}$

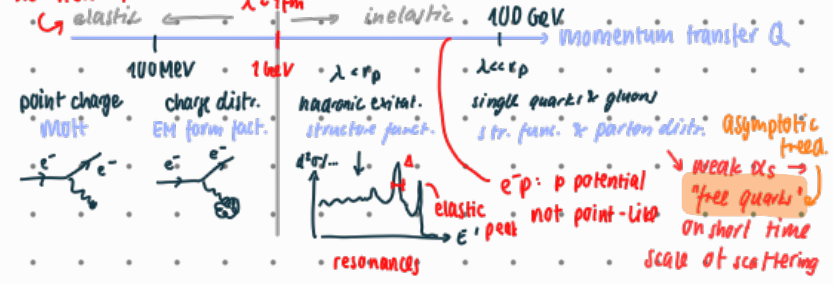
Scattering experiments:



background dominates (strong)
 $\rightarrow$ only $\bar{u}d \rightarrow W^- \rightarrow e^- \bar{\nu}_e$ & $u\bar{u} \rightarrow Z \rightarrow e^+ e^-$
usable to detect W



always look at given energy when interpreting $d\sigma/dE(Q)$
no new particles



elastic	inelastic
dynamics: Form factors (G_M, G_E) Invariant mass of the resona. $W = M: 2M\nu - Q^2 = 0$ ($r = \frac{P_q}{M}$) given E_e: 1 free parameter θ $\sigma = f(\theta)$	structure functions (W_1, W_2) $W > M: 2M\nu - Q^2 > 0$ $\sigma = f(E', \theta)$ or $f(Q^2, \nu)$ Since $\Delta E \cdot \Delta t \sim \hbar$, we can estimate lifetime of a resonance $\tau \sim \frac{\hbar}{\Gamma}$ (Γ: FWHM of $\sigma_{tot}(E)$)

De Broglie: $\lambda = \frac{h}{p} = \frac{hc}{\sqrt{2Mc^2 + T^2}}$

Bragg condition: $n\lambda = 2d \sin \theta$ Angle of diffraction $\propto \lambda$
difference in path length

Electron diffraction & Rutherford: Incoming particle wave is coherently split (diffracted) on scattering centers. Look at interf. in outgoing waves. $\odot$: sensitive

Resolution of a particle: $\Delta x = \frac{\lambda}{2\pi} = \frac{h}{p\lambda} \rightarrow p \approx \frac{hc}{\Delta x} \sim 200 \text{ MeV fm}$ $\rightarrow$ Momenta required to resolve structure:

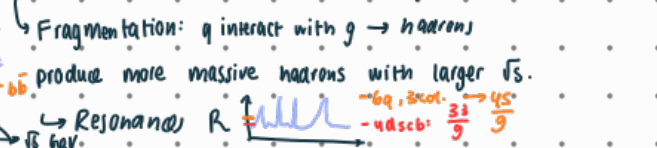
Quark distribution: $F_2^{ep}(x) \propto \frac{1}{9} (d_v^2 + \bar{d}_s + \bar{s}_s) + \frac{4}{9} (u_v^2 + \bar{u}_s + \bar{s}_s)$

Atomic nucleus: 10-100 MeV/c $\odot \sqrt{s} \approx 2M_p$ or $2M_q$
proton: $\sim 100 \text{ MeV/c}$ & quarks: several GeV/c
 $e^- e^-$ Annihilation: $e^- e^- \rightarrow \mu^+ \mu^- / q \bar{q}$ ($\sqrt{s} \ll M_Z c^2$)

$\hookrightarrow$ in p. for ep scattering, valence & sea

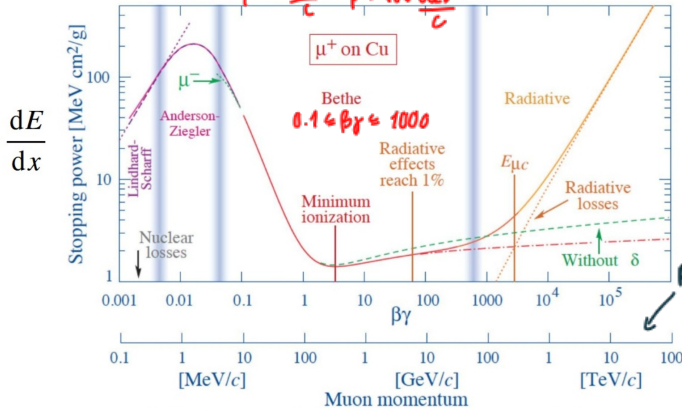
q appear equally often in n & p & e^- scatter. $F_2(x) = \frac{F_2^{ep} + F_2^{en}}{2} = \sum_{f=u,d} \sum_{i=1}^3 (q_f(x) + \bar{q}_f(x))$
Fragmentation: q interact with $g \rightarrow$ hadrons

Inelastic scattering between e^- & $p \leftrightarrow$ elastic scattering e^- & partons
 $\odot$: e^- have a larger λ for similar E_{tot} than other particles. x are point-like



Interaction with matter

low-energy ionization radiative



log-growth: E-Field increases from Lorentz factor, but medium polarization → field shielding
high energies: Hadronic showers for quarks

Energy loss of particles in matter:

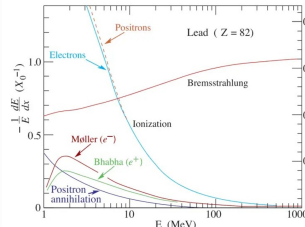
photons:

$$I(x) = I_0 e^{-\mu x}$$

- PE: $\mu_{PE} \sim \frac{Z^4}{E^{3.5}}$
- CS: $\mu_{CS} \sim \frac{Z^2}{E}$
- PP: $\mu_{PP} \sim \frac{Z^2}{E}$

Mass attenuation length / mean free path: $\lambda = \frac{1}{\mu}$
 $= N_A \frac{\sigma}{A}$

same E_{kin}
→ lighter part. faster & loses less energy



charged particles:

$$\frac{dE}{dx} = \left(\frac{dE}{dx} \right)_{ion} + \left(\frac{dE}{dx} \right)_{rad}$$

ionization losses, Bethe-Bloch
radiative losses $\sim Z^2 E / m^2$
imp. for e^+e^- heavier: only ultra-relat.

→ e^- liberated by ionization, have large E

Bethe-Bloch: $-\frac{dE}{dx} = \frac{4\pi N_A r_e^2 Z^2}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2mc^2 \beta^2 \gamma^2 W_{max}}{I^2} \right) - \beta^2 \delta(\beta\gamma) \right]$

losses depend only on v , not mass. lighter particles are faster than heavier nuclei.
Valid only for heavy charged particles (no e^-)

$K = 4\pi N_A r_e^2 m_e c^2 Z^2 \approx 0.31 \text{ MeV g}^{-1} \text{ cm}^2$
 $W_{max} = \frac{2mc^2 \beta^2 \gamma^2}{1 + 2\gamma m_e / M + (m_e / M)^2}$ max energy transfer in single collision, $M \gg m$
Range of part: $R = \int \frac{dE}{dE/dx}$
High energy limit: $-dE/dx = CZ^2/\beta^2$, $C \approx 1.7 \text{ MeV cm}^2$

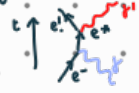
Photoelectric effect: γ absorbed, orbital e^- kicked out. Energy passed to e^- & nuclear recoil. Dominant at low energies. $E_\gamma = h\nu$, $E_e = E_\gamma - E_B$



$\mu_{PE} \sim \frac{Z^4}{E^{3.5}}$ $3 < n < 5$ * Binding energy of e^- in K-shell $\sigma \propto E^{-3}$ with discontinuities for ionization of diff. atomic levels

Compton Scattering: orbital e^- kicked out, γ scattered with lower energy. Dominant at intermediate energies. $E_\gamma = h\nu$, $E_e = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2} (1 - \cos \theta)}$

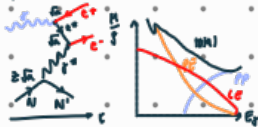
dominant for $1 \text{ keV} \leq E \leq 500 \text{ MeV}$



often: e^- initially at rest. $e^- + \gamma \rightarrow e^- + \gamma'$ $\leftrightarrow p_e + p_\gamma = p'_e + p'_\gamma$ $(p'_e)^2 = (p_e + p_\gamma + p'_\gamma)^2$
 $\leftrightarrow m_e^2 = m_e^2 + m_e(p_\gamma + p'_\gamma) - p_\gamma p'_\gamma (1 - \cos \theta)$ with $p_\gamma = (|\vec{p}_\gamma|, \vec{p}_\gamma)$ & $A \cdot B = (a_0 b_0 - (a_1 b_1 + a_2 b_2 + a_3 b_3))$
goes up to 'Compton edge'. Max. energy deposit by γ through $180^\circ \rightarrow$ backscattering. Wave length shift: $\Delta \lambda = \frac{h}{m_e c} (1 - \cos \theta)$
→ $\Delta E = m_e c^2 / 2$

Pair Production: e^+e^- pair created in field of nucleus or e^- . Min. energy: $E_\gamma \geq 2m_e c^2$. $E_e \geq m_e c^2$, $E_\gamma = h\nu$. $\mu_{PP} \sim Z^2 E_\gamma$. Dominant at high energies

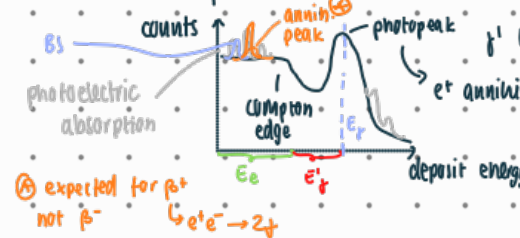
dominant for $E \geq 1 \text{ MeV}$



only in presence of external nucleus/atomic e^- , similar to BS → EM showers. Cross section: raises above threshold but saturates at large E_γ (screening effects)

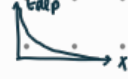
γ : PP, e^- : BS → # of part. doubles every X_0 . Absorption coeff: $\mu = n\sigma = \rho N_A \sigma / A$

Gamma spectroscopy: β decays accomp. by γ decays. Interaction of γ -radiation in matter. Most photons are absorbed when they interact in material. processes that can happen during detection: Compton scattering, Backscattering, Photoel. absorption, Annihilation, Pair production



γ' has min. energy for $\theta = 180^\circ$. Backscatter energy for $E_\gamma \approx m_e c^2$: $E_{\gamma'} \approx 200 \text{ keV}$. CE & Photopeak always together. BS peak is a direct consequence of CS in the surrounding material.

No continuous energy loss. $I(x) = I_0 e^{-\mu x}$ attenuation coeff: $\mu = \mu_{PE} + \mu_{CS} + \mu_{PP}$



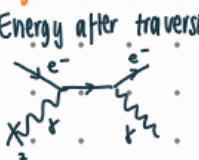
Synchrotron Radiation: main source of energy loss for ultra-relativ. particles. Energy loss: $\frac{dE}{dt} = P = \frac{e^2}{6\pi \epsilon_0 c^3} \left(\gamma^2 \frac{v^2}{R} \right)^2 \sim \frac{E^4}{m^4 R^2}$

* Bremsstrahlung. γ emission in QCD!

→ jets = proof of gluons

angular distr. → spin & η !

Energy after traversing x: $E(x) = E_0 e^{-x/X_0}$ & $X_0 \approx 180 \frac{A}{Z^2} \text{ g cm}^{-2}$ (A : #n+p, Z : #im PS) → $X_0 = \frac{1}{70 N_A Z^2 r_e^2 \ln(\frac{183}{Z})}$



Critical energy: energy loss via collisions = energy loss via radiation: $\frac{dE}{dx} /_{ion.} = \frac{dE}{dx} /_{B.S.}$ & $E_c = \frac{710}{Z + 1.2} \text{ MeV}$ (gas) (solid)

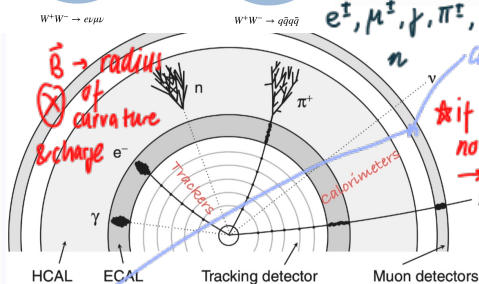
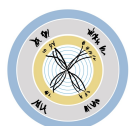
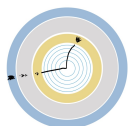
above E_c , Bremsstrahlung dominates
→ e^- + field/nucleus → $e^- + \gamma$

σ_{tot} : e^-e^+ annih., ioniz., bremsstr., e^-e^+ scattering

Detectors (measure diff. cross sect.: $\frac{d\sigma}{d\Omega dE'}$)

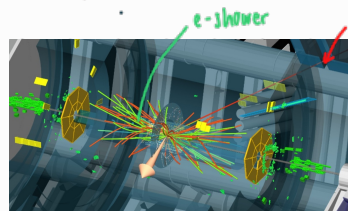
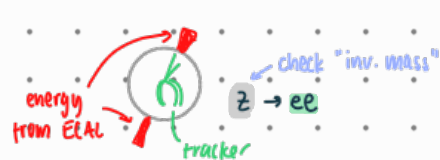
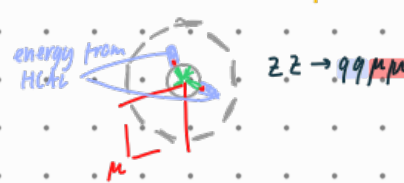
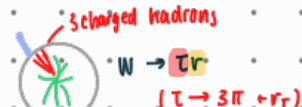
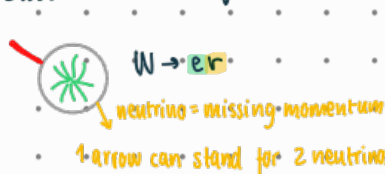
Resonance: detection of short-lived particles. An unstable particle can decay into diff. "channels" $\rightarrow$ Branching ratio $B_r = \frac{\Gamma_r}{\Gamma_{tot}}$

Δ^{++} : uuu, $|\frac{3}{2}, \frac{3}{2}\rangle$, Δ^+ : uud $|\frac{3}{2}, \frac{1}{2}\rangle$
 Δ^0 : udd, $|\frac{3}{2}, -\frac{1}{2}\rangle$, Δ^- : ddd $|\frac{3}{2}, -\frac{3}{2}\rangle$



- e^+ : leaves track + EM energy deposit (edep)
- γ : no track but EM edep, EM shower in EDEP, no track in inner tracker
- $\mu^\pm$: track, small calo energy deposits, penetrating
- ν : not detected (need special detectors) appear
- hadrons: track (if charged) + HCAL edep
- quarks: seen as jets of hadrons

particles where $v \approx c$ suffer from time dilation: $\tau_{lab} \gg \tau_{rest} \rightarrow$ leave long track in Bubble Chamber (charged p. in B-Field.) c^- small R. vertex = neutral part. decay



Scintillators: Noble gases & liquids, inorganic crystals (largest light yield, precis. meas. of energetic γ or calorimetry)

Cherenkov: atmospheric ν detection: ν_e, ν_μ hit water $\rightarrow e, \mu$ if large $E_{kin} \rightarrow$ light cone $\rightarrow$ ring

$v > \frac{c}{n} \rightarrow$ radiation is emitted in a cone with angle $\cos \theta = \frac{1}{n\beta} \rightarrow$ mass momentum inequality $mc < (n^2 - 1)^{1/2} p$ $\times$ charged

MWPC: array of wires @ high voltage, plates at ground. **TPC:** low field inside volume to not cause e^- shower. Measure arrival time of e^- .

Trackers: measurement of trajectories to identify vertex (origin of particles). Low-density materials to minimize energy deposit. +B-Field: get trajectory, charge & p .

Calorimeters: measures particles energy ($\gamma, e^-,$ hadrons) $\rightarrow$ showers. High-density materials.

Scintillators: Emission of γ by excited atoms. Used for tracking, calorimetry & timing: Noble gases, inorganic crystals

Cloud chamber: thick trace $\rightarrow$ more initial ionization $\rightarrow$ more $\frac{dE}{dx}$ (α : thick, short, β : thin & long, low $\frac{dE}{dx}$) forks & kinks: Interaction & decay

Photon detectors: - photomultiplier tube (PMT): $\gamma \rightarrow e^-$ (photoeffect). e^- is acc. & multipl. $\odot$: sensitive to single photons. $\odot$: sensitive to B-Field. $\odot$: low granularity. - silicon photomultipliers: reversly biased p-n junction. $\odot$: no sens. to B-Field, high gran. $\odot$: small area, noise

Experiments with fixed target: • Scattering: elastic: e^-e^- Coulomb field: proton is not pointlike! inelastic: p^+ has pointlike objects inside $\rightarrow$ quarks • Secondary beams: Rare isotope production, Neutrino experiments, Particle at rest: Decays & bound states

Collider experiments: e^-/p collisions. New particle discoveries (appear as a resonance): using # of events as a function of inv. mass of collision, this appears as a peak in the production cross section



e^+e^- more precise than pp: $\rightarrow$ have no structure $\rightarrow$ Wider resonance curve

Accelerators: mostly used for industry, health & energy, prepares partic. with E, p & $\vec{p}$ linear pre-accelerator using RF cavities to generate high freq. asc. E-Fields. Natural sources are cosmic rays, natural radiation $\alpha, \beta, \gamma \rightarrow$ lack energy or intensity & luminosity. Hard to gain control. Needed to produce precise initial states with variable E , at high intensities possible to produce 19^+ ≈ 2 Mw C (weak stronger than Em).

accelerating charged particles: they travel through a fixed high voltage $U \rightarrow$ gains energy $\Delta E = eU = q \int_{r_1}^{r_2} (\vec{v} \times \vec{B} + \vec{E}) \cdot d\vec{r} = qU$ technical limit for E-Field: $E \propto \frac{10MV}{m}$ solution: alternating fields: Linear accelerators. Energy after n acc. gaps: $E = nqV_{max} \sin \phi_s$. For medium E. High E need superconducting (resonant to injected RF freq. $\rightarrow$ Final energy limited by length & peak RF voltage) length dep. on q/m , $\frac{\Delta E}{\Delta L} = \frac{2qV_{max}}{\lambda_{RF} \beta_{tr}}$

Cyclotrons: $\omega = \frac{qB}{m}$, Output energy: $E = \frac{q^2 B^2 R^2}{2m}$ limited by relativistic effects, orbit: Lorentz force $\rightarrow \Delta E \propto \frac{1}{m\gamma}$ **Synchrotron:** signal has to move faster than particles, magnets $\rightarrow$ limitation $(\Delta E = \frac{4\pi e^2 \beta^2 \gamma^4}{3R})$, $\omega = \frac{1}{R} \frac{Be}{m}$ energy loss

Isospin not conserved by weak interact. $\frac{1}{2}(n_u - n_d)$

states are described by I (tot. isospin) & I_3 (EV of z-projection). Each I_3 state specifies certain flavor of multiplet. Spin & Isospin describe

$|u\rangle = |I, I_3\rangle = |\frac{1}{2}, \frac{1}{2}\rangle = \uparrow$ $\bar{d} = -|\frac{1}{2}, \frac{1}{2}\rangle$ $c: I_3 = 1$ $b: +1$ ($\bar{q}$: all signs flip)
 $|d\rangle = |I, I_3\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle = \downarrow$ $\bar{u} = |\frac{1}{2}, -\frac{1}{2}\rangle$ $s: -1$ $b: -1$

lightest hadrons only

calculate: $|I^A, I_3^A\rangle |I^B, I_3^B\rangle = |I, I_3\rangle$ with: $\begin{cases} I \in \{ |I^A - I^B|, |I^A + I^B| \} \text{ integer steps } (\frac{1}{2}, 1, \dots \text{ or } 1, 2, \dots) \\ I_3 = I_3^A + I_3^B \end{cases}$

$\Lambda^0: |0, 0\rangle = |0, 0\rangle \oplus |1, I_3\rangle = |1, I_3\rangle$
 $K^0: |\frac{1}{2}, \frac{1}{2}\rangle, K^+ = |\frac{1}{2}, \frac{1}{2}\rangle$
 $\Sigma: I = 1, \Delta: I = \frac{3}{2}$

$\epsilon_x: |1, 1\rangle = (-|\frac{1}{2}, \frac{1}{2}\rangle, -|\frac{1}{2}, \frac{1}{2}\rangle) = -\bar{u}\bar{d} = \pi^+$, $|1, 0\rangle = \frac{1}{\sqrt{2}}(|\frac{1}{2}, \frac{1}{2}\rangle - |\frac{1}{2}, -\frac{1}{2}\rangle) = (\bar{u}\bar{u} - \bar{d}\bar{d})/\sqrt{2} = \pi^0$ & $|1, -1\rangle = \bar{d}\bar{d} = \pi^-$

consider only $\frac{3}{2}$ & (has $I = \frac{3}{2}$)

p & n doublet: $I = \frac{1}{2}, I_3 = -\frac{1}{2}, \frac{1}{2}$ singlet: $|0, 0\rangle = \frac{1}{\sqrt{2}}(|\frac{1}{2}, \frac{1}{2}\rangle - |\frac{1}{2}, -\frac{1}{2}\rangle) = (\bar{u}\bar{u} + \bar{d}\bar{d})/\sqrt{2} = \eta^0$

$\pi^+, \pi^0, \pi^-: I = 1, I_3 = -1, 0, +1$
 always for neutr. meson
 total spin: $\frac{3}{2} \rightarrow$ high mass
 $\Delta^{++}: u^+u^+u^+, \Delta^+: u^+u^+d, \Delta^0: u^+ud, \Delta^-: udd$
 connection to el. charge: $Q = e(I_3 + \frac{1}{2})$
 parity (+) $\rightarrow$ lowest excited state

Ex. $\pi^+ p \rightarrow \Delta^+ \rightarrow \pi^+ p$ (1)
 $\pi^- p \rightarrow \Delta^0 \rightarrow \pi^0 n$ (2)
 $\sigma_1 = \frac{|\langle \pi^+ p | H_S | \pi^+ p \rangle|^2}{|\langle \pi^0 n | H_S | \pi^- p \rangle|^2}$
 $\sigma_2 = \frac{|\langle \pi^0 n | H_S | \pi^- p \rangle|^2}{|\langle \pi^+ p | H_S | \pi^- p \rangle|^2}$
 $= \frac{1 \cdot 1^2}{|\frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}|^2} = 9$
 Assume H_S indep. of I_3 & isospin symmetry

Clebsch-Gordan: 1. What table? $1 \times \frac{1}{2}, \frac{3}{2} \times \frac{1}{2} \dots \rightarrow$ Isospin for each particle: $|I_1, I_{1z}\rangle \rightarrow I_1 \times I_2$
 2. tot $\in \{|I_1 - I_2|, |I_1 + I_2|\}$ in full steps $\rightarrow$ we have values of $I_{tot} \rightarrow$ need coeff. $\rightarrow$ not nec. all possible!

1	1/2	3/2	3/2	1/2	1/2
1	1/2	1	1/2	1/2	1/2
0	-1/2	1/3	2/3	1/3	1/2
0	1/2	2/3	-1/3	1/3	-1/2
0	-1/2	2/3	1/3	1/3	3/2
-1	1/2	1/3	-2/3	-1/3	-3/2
-1	-1/2	-1/3	2/3	-1/3	1

$\pi^- + p = |\pi^-, p\rangle = |1, -1\rangle \otimes |\frac{1}{2}, \frac{1}{2}\rangle \rightarrow M = m_1 + m_2 = -\frac{1}{2}$
 $\Rightarrow = \frac{1}{\sqrt{3}}|\frac{3}{2}, -\frac{1}{2}\rangle - \sqrt{\frac{2}{3}}|\frac{1}{2}, -\frac{1}{2}\rangle \rightarrow$ can show $\pi^0 \rightarrow \pi^+ + \pi^-$ (not $2\pi^0$)

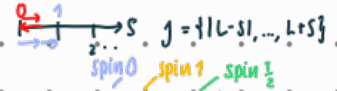
the cross section $\sigma \propto \langle I', I'_3 | H_{strong} | I, I_3 \rangle^2 \xrightarrow{[H_S, I_3] = 0} \sigma \propto \langle I', I'_3 | I, I_3 \rangle^2$

Color: r, g, b. $\bar{q}$ have anti-red, etc. qq can't exist (color confinement)
 Hadrons are color neutral: Baryons: $|qqq\rangle = \frac{1}{\sqrt{6}}(|rgb\rangle - \dots)$ antisymm. to color change
 Mesons: $|q\bar{q}\rangle = \frac{1}{\sqrt{3}}(|r\bar{r}\rangle + \dots)$ symmetric to color change
 gluons never change the quark's flavour. Carry 2 colors: $3 \otimes \bar{3} = 8 \oplus 1$ they carry color $\rightarrow$ can interact with themselves, 8 gluons

Quark confinement: not possible to isolate single quarks (not colorless) $\rightarrow$ strong interaction even though $m_q = 0$ is short-range (q bound to $\bar{q}$ in hadron)

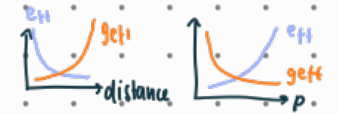
R indep. of α needed for Pauli exclusion principle. Experimental confirm: collide $e^+e^- \rightarrow$ annihilation via γ^* . $\sigma(e^+e^- \rightarrow \mu^+\mu^-) \propto \alpha^2$ & $\sigma(e^+e^- \rightarrow q\bar{q})$
 (coupled in same way) $\propto \sum_f e_f^2 \alpha^2 \rightarrow R = \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = N_c \sum_f e_f^2 \rightarrow$ Assuming 6 quarks with $2f \in \{\frac{1}{3}, -\frac{1}{3}\}$ we get $R = \frac{11}{9} N_c$ & measure $N_c = 3$
 fractional charge of q, norm. el. charge R increases if $E_{CM} \uparrow \rightarrow$ more (x heavier) q can be produced ($\exists$ more processes)

Baryon No.: $q: \frac{1}{3}, \bar{q}: -\frac{1}{3}, \frac{q}{e} = I_3 + \frac{(S+B)}{2}$ If conserved: balance between matter & anti-matter. Sim expect.: $\frac{n_B - n_{\bar{B}}}{n_\gamma} \sim 10^{-18}$ vs. obs.: $\sim 10^{-10}$
 necessary to explain stability of proton. Else $p \rightarrow \pi^0 + e^+$ would be possible.

Angular momentum: $|meson\rangle = |q\bar{q}\rangle$ 
 looking at decays (e.g. $\pi^0 \rightarrow \gamma\gamma$). Tot. ang. mom.: $J = S + L = 1 \rightarrow$ possible L-S combinations for final state?
 Since $J = |L - S|, \dots, |L + S| \rightarrow$ look for comb. s.t. J is in range $\rightarrow J = 1, L = 1$
 $\hookrightarrow$ Antisymmetry for fermions. Can't occupy same quantum state $\rightarrow \Psi$ antisymm. (Bosons: symm.)

Higgs field: min. Energy state occurs at non-zero field values. Groundstate & symmetry. Scalar field. Weinberg angle: Choose vacuum state by rot.

Shift vacuum value with $\Phi_0 \rightarrow$ const. term generates a mass term for particles ($\odot$ fields have mass too)
 Electroweak unification & Higgs mechanism $\rightarrow$ Weak isospin & hypercharge.
 $\begin{pmatrix} |1\rangle \\ |2\rangle \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} |B\rangle \\ |W\rangle \end{pmatrix}$ & $\frac{m_W}{m_Z} = \cos\theta_W$; theory based on obs. gauge symmetries



$\rightarrow$ resonant peak
 H-production: gluon-gluon fusion, vector-boson fusion, Higgs-strahlung, $t\bar{t}$ -fusion

Changes under C, P and T transformations:
C: electric charge, baryon number, lepton number, strangeness, charm, bottomness, topness, magnetic dipole moment, electric potential, electric field, helicity for MASSLESS PARTICLES

P: Position, Momentum, Velocity, Acceleration, Force, Electric Field, Electric Field, Electric Potential
T: Time, Velocity, Momentum, Angular Momentum, Magnetic Field

Figure 3: Wu experiment decay schema

$$\begin{aligned} \pi^0 &\rightarrow \gamma + \gamma: C_\pi = C(\gamma_\gamma) = (-1)^2 = 1 \\ 2^3S_1: C &= (-1)^{L+S} = (-1)^{0+1} = -1 \\ D: &(-1)^L(-1)^S(1+1) = -1 \end{aligned}$$

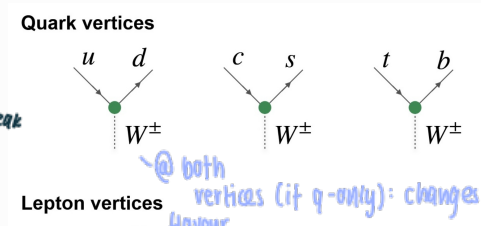
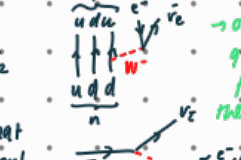
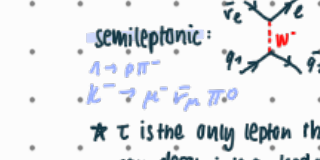
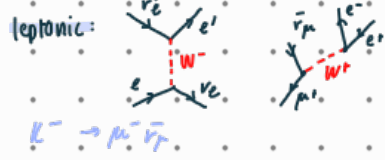
violated
 symmetry (time reversal).
 maximally violated part of SM

$\tau \sim 10^{-13}s \rightarrow$ long life $\rightarrow$ much weaker than EM only at low E due to large W mass

Weak interaction

large mass $\rightarrow$ short lifetime

- Z : conserves flavour
- $W^\pm$: changes flavour
 - leptons: doesn't change generation. conserves e, μ & τ number. g_W
 - quarks: can change generation.

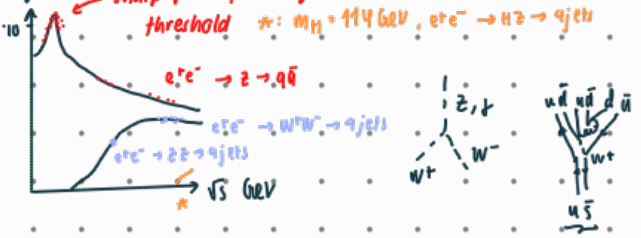


At low energies: $q^2_{max} \ll m_W^2$ (W boson "very virtual"). Effective field: $r_0 \propto \frac{1}{m_W}$ (tiny range), only at low E much weaker than EM interaction $\alpha_{EM} \approx \alpha_W$

At high energies: using G_F is insuff. We can have: $\nu_\mu + e^- \rightarrow \mu^- + \nu_e$ for $E \rightarrow \infty$: $\sigma \propto G_F^2 m_W^2 \propto \frac{g_W^4}{m_W^2}$ for $s = -q^2 = Q^2 \rightarrow \infty$

Lepton number: $L = N(e) - N(\bar{e}) + N(\nu_e) - N(\bar{\nu}_e)$. Ex: $\mu \rightarrow e \gamma$ not possible $\rightarrow$ need $\nu \bar{\nu}$ $\rightarrow$ "kink" in tracks

- top mass too large: $W^+ \not\rightarrow t \bar{b}$, $Z \not\rightarrow t \bar{t}$
- W & Z production at LEP:
 - $e^+e^- \rightarrow Z$: 2 jets
 - $e^+e^- \rightarrow WW$: 4 jets
 - $e^+e^- \rightarrow ZZ$: 4 jets, ee



- $W^\pm$: nuclear decay: $^{60}_{22}Co \rightarrow ^{60}_{22}Ni + e^- + \bar{\nu}_e$
- $n^0 \rightarrow p^+ + e^- + \bar{\nu}_e$ β -particle: very fast e^- or e^+
- $W^- \rightarrow e^- + \bar{\nu}_e$ decays immediately

Discovery of W & Z bosons: High energy col. of hadrons with $E_{cm} = (p_1 + p_2)^2 \rightarrow$ resonance (peak in σ) when $\sqrt{s} \approx m_W$ or m_Z . Z decays to $\nu \bar{\nu}$, W to $q \bar{q}$, $g \bar{g}$ or $l \bar{l}$.

- π^+ & π^- lightest hadrons $\rightarrow$ decay mainly to leptons. $\pi^+ \rightarrow e^+ \nu_e \pi^0$ & $\pi^- \rightarrow e^- \bar{\nu}_e \pi^0$ possible but unlikely: small branching ratio (3 particle final state $\rightarrow$ small phase space)
- $\pi^- \rightarrow \mu^- \bar{\nu}_\mu$ more likely than $\pi^- \rightarrow e^- \bar{\nu}_e$ since μ has a larger righthanded ($h=1$) fraction than e : $W(h=1) = \frac{1}{2}(1-\theta)$

angular mom. conserved: π : spin 0 $\rightarrow$ spin of μ, ν point away from each other $\rightarrow$ Z has $h=-1 \rightarrow$ needs $+1$ to interact.

"heavier products inherit less velocity $\rightarrow$ bigger space of accessible frames in which helicity can be flipped" $\mu^- \leftarrow \bullet \rightarrow \bar{\nu}_\mu$ W distinguishes chirality, not helicity ($h=+1$ suppressed)

π^- has 0-spin $\rightarrow$ spin of $\bar{\nu}_\mu$ & μ^- are opposite, W only couples RH chiral $\bar{\nu}_\mu \rightarrow$ RH helicity $\rightarrow$ conserve ang. mom. $\rightarrow$ μ has RH helicity

- W-boson couples only to left-handed particles & right-handed anti-particles $\rightarrow$ only left-handed particles decay
- $\sin^2 \theta_W = 1 - \frac{M_W^2}{M_Z^2} \rightarrow$ measured M_W, M_Z determine θ_W .
- Ex: $\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$. e^- have most E_{kin} if they are $\parallel$ to μ , ν_μ & $\bar{\nu}_e$ opposite dir. $\rightarrow e^-$ right (as muon was)
- A particle with non-zero strangeness can decay $\rightarrow$ zero-str. only via weak interaction
- Z^0 : neutral current, is its own antiparticle. Only transfers spin and/or momentum. For neutrino scattering

e^- -capture: weak process. Cross sect. increases with $Z \uparrow$ $e^- + {}^A_Z P \rightarrow {}^A_{Z-1} D + \nu$ (inner shell, e^-)

relates weak eigenstates with mass eigenstates of quarks

CKM matrix: $g_W^u \leq g_W^u \leq g_W^d$ $\begin{pmatrix} |d^1\rangle \\ |s^1\rangle \\ |b^1\rangle \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} |d\rangle \\ |s\rangle \\ |b\rangle \end{pmatrix}$ where $|V_{qq'}|^2$ is the probability for the transition $q \rightarrow q'$

unitary

0.97370 ± 0.00014	0.2245 ± 0.0008	0.00382 ± 0.00024
0.221 ± 0.004	0.987 ± 0.011	0.0410 ± 0.0014
0.0080 ± 0.0003	0.0388 ± 0.0011	1.013 ± 0.030

estimate the branching ratio of Z decays: Assume $\Gamma(a \rightarrow bc) \propto |M(a \rightarrow bc)|^2 \cdot g(a \rightarrow bc)$

all terms but effective coupling constants cancel: (if same generation: $V_{ud} \approx V_{cs} \approx \cos \theta_C$, else: $V_{us} \approx -V_{cd} \approx \sin \theta_C$)

Sargent Rule: transition rate $\Gamma \propto E_0^5$. E_0 : available energy in decay, depends only on phase space: $g \propto E_0^5 = (m_a - m_b - m_c)^5$

Ex: $\sigma(K^+ \rightarrow \mu^+ \nu_\mu) \propto |M|^2 \propto (g_W \sin \theta_C)^2 \cdot (W^\pm \text{-propagator}) \cdot \theta_W^2 \propto g_W^4 \sin^2 \theta_C$ kann auch $(\theta_C)^4$ sein, bei 2 q-changes

Width of Z-resonance: $\Gamma_{tot} = \Gamma_e + \Gamma_\mu + \Gamma_\tau + \Gamma_g + N_f \Gamma_f$. $\sigma(\sqrt{s} \approx M_Z) \propto \frac{\Gamma_e}{\Gamma_{tot}} \frac{\Gamma_f}{\Gamma_{tot}}$ $\rightarrow$ solve for N_f (≈ 3), only works for $m_f \ll \frac{M_Z}{2}$

weak charge universality: $\Gamma_{\tau \rightarrow \mu} \approx \Gamma_{\tau \rightarrow e} \approx \Gamma_{\tau \rightarrow \nu_\tau} \approx 3 \Gamma_{\tau \rightarrow e}$ $\Gamma_{\tau \rightarrow b} \propto m_b^5$, $\tau_\tau = \frac{\hbar}{\Gamma_{tot}} \approx 2.9 \cdot 10^{-13}s$ ($\tau \rightarrow \pi$ 3x more likely than $\tau \rightarrow e$)

Nuclear Physics

giant resonances: Result from collective movement of sev. n & p around COM. $\Delta E \uparrow$ if rot. quant. No. $\uparrow$, nuclear mass $\uparrow$

Fermi gas model: Fermi energy indep. from # of nucleons. Assume $m_N = m_p$. Non-relativ. limit: $E_F = \frac{p_F^2}{2m}$, $N = Z = A/2$, const. density: $R = r_0 A^{1/3}$

all at ground state. Structure of nuclei, explains measured shape of potential & $\sim$ const. binding energy

$\rightarrow$ min. Nucleus E_{kin} if $N \neq P$ (small asymmetry) $\rightarrow$ large $Z \rightarrow$ more Coulomb $\rightarrow$ more N to counteract.

large, stable nuclei: $N > P$ (fusion frees more energy than fission)

Kurie plot comes from Sargent's rule:

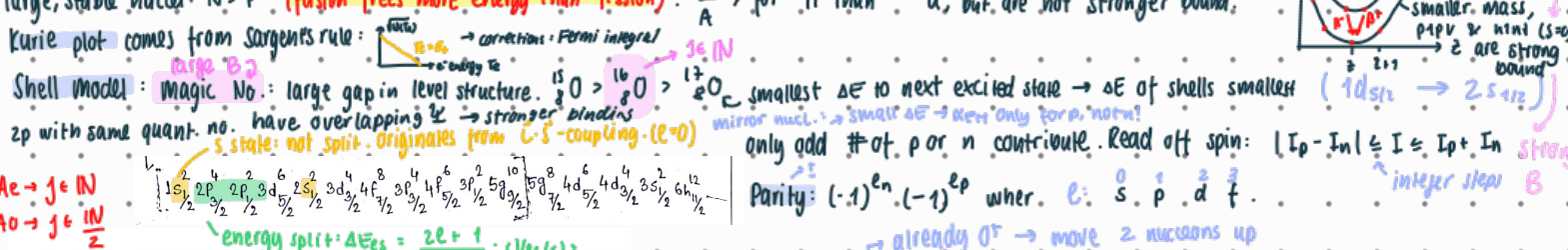
Shell model: Magic No.: large gap in level structure. $2, 8, 20, 28, 50, 82, 126$

2p with same quant. no. have overlapping $\rightarrow$ stronger binding

s. state: not split. Originates from C-S coupling ($L=0$)

only odd # of p or n contribute. Read off spin: $|I_p - I_n| \leq I \leq I_p + I_n$

Parity: $(-1)^{e_n} (-1)^{e_p}$ where $e: s, p, d, f$



Decay Type: α, β, γ

Radiation Emitted: α, β, γ

Generic Equation: $A_Z X \rightarrow A'_Z X' + A''_Z X''$

instabilities: α easy to shield

Alpha decay: $E_\alpha = E_i - E_f$

conserves C & Parity

classically not possible (Coulomb barrier $\rightarrow$ tunneling)

Beta decay: $A_Z X \rightarrow A_{Z+1} Y + e^- + \bar{\nu}_e$

outside the nucleus

needs to satisfy: $m_{parent} - m_{daughter} - m_e > 0$

(energy balance). favorite for neutron-rich isotopes, $N > Z$

Weizsäcker: $M \propto Z^2$

Beta+ decay: $A_Z X \rightarrow A_{Z-1} Y + e^+ + \nu_e$

proton in nucleus (E_{max}, e^+)

energy diff.: $E_0 = m_{parent} - m_{daughter} - m_e - m_{\nu}$ (keV)

(after α -decay) nuclear deexcitation, MeV range

Gamma emission: $E_\gamma = E_i - E_f$

emit γ (C conserv.), long τ : ang. mom. change, E_γ for work

$\lambda_\gamma \gg$ nuclear size

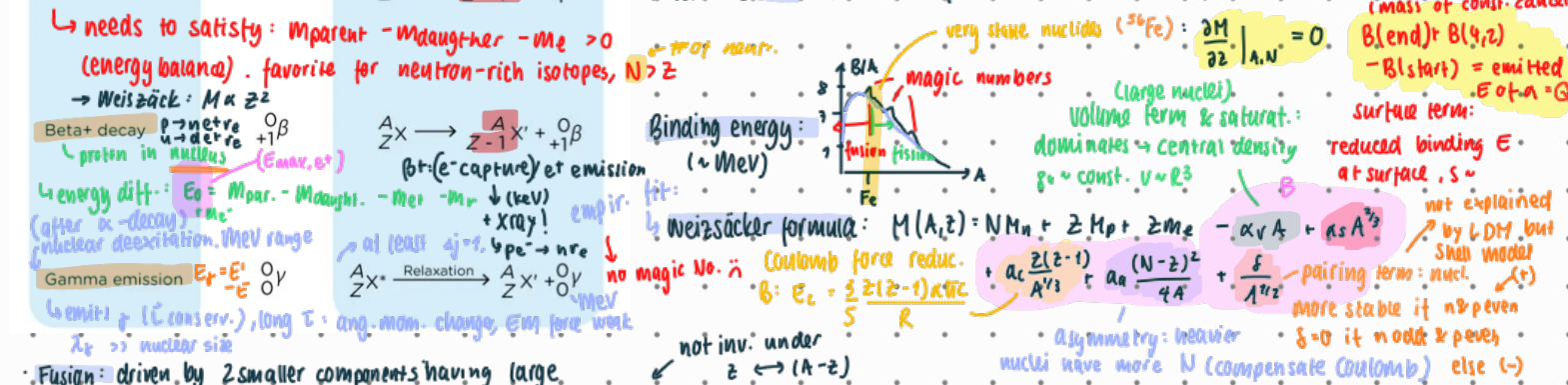
Fusion: driven by 2 smaller components having large surface energy compared to vol., decreases after fusion. Coulomb E increases (smaller effect)

Surface energy: compared to vol., decreases after fusion. Coulomb E increases (smaller effect)

H $\rightarrow$ He Fusion d+t $\rightarrow$ He + n, hindered by, via tunneling through Coulomb barrier.

neutron capture (higher prob., no Coulomb $\rightarrow$ excite nucl. to deform. $\rightarrow$ fission)

Fission: energy in Coulomb term decreases, surface E increases, nucleus $\rightarrow$ 2+ daughters. Reactors & bombs, U-235, Pl-239



critical mass: needed to sustain chain reaction (geometry)

water: moderates prompt & delayed neutrons: $T \rightarrow$ $g \rightarrow$ less moderation, cools down to get E_{therm} .

$Q = B(A_1, Z_1) + B(A_2, Z_2) - B(A_3, Z_3)$

nucleus deforms, eg: $200X \rightarrow 100Y + 100Y$

$-B(A, Z) > 0$, lifetime falls as Z^2/A goes up, very rate. α is favorable (high $B \rightarrow E_\alpha$ higher)

higher probability if n is thermal: $E_{kin} \sim 25 MeV$ & nuclei n : $E_{kin} \sim 10 MeV$ $\rightarrow$ tunneling $\rightarrow$ escape nucl.

Neutron absorption: excite nuclei: $\sigma \propto \lambda^2 \pi$, $\lambda \propto \frac{1}{v} \sim \frac{1}{\sqrt{E}}$ $\rightarrow$ Induced fission of ^{235}U (^{238}U has highest abundance but can't absorb n)

Chain reaction: neutron multiplication factor: $k=1$: sustainable, $k<1$: sub critical, $k>1$: superc. $\rightarrow$ Fission reactors: $k \sim 2.5 \rightarrow$ ^{235}U is rare, ^{238}U no fission

Nucleon-nucleon scattering: at low E (E for π -prod.), elastic, assume point-like part.

don't neglect spin & isospin. Subtract Coulomb cross sections (known from e.p scattering)

Yukawa potential: $V = g \frac{e^{-m_r r}}{r}$ (exchange of mesons with mass, m) $\rightarrow$ predicts pion masses, $\sim 140 MeV$

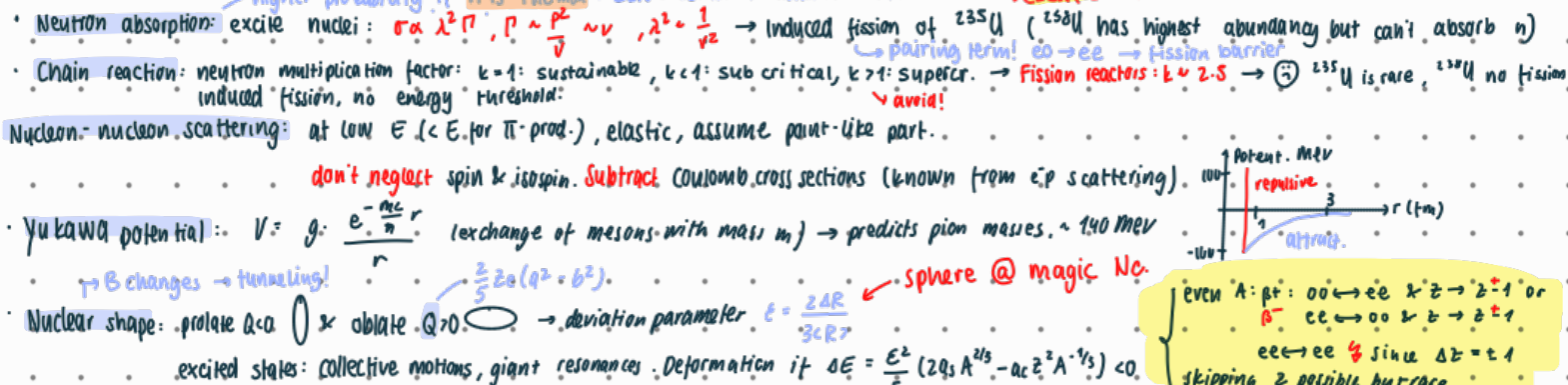
$\rightarrow B$ changes $\rightarrow$ tunneling!

Nuclear shape: prolate $Q < 0$ & oblate $Q > 0$ $\rightarrow$ deviation parameter $\epsilon = \frac{2AR}{3C R^2}$

excited states: collective motions, giant resonances. Deformation if $\Delta E = \frac{E^2}{5} (2Q_0 A^{2/3} - a_2 Z^2 A^{-1/3}) < 0$

Gamow model: for α -decays. Explains why E spectrum is discrete & the Geiger-Nuttall rule ($\tau = \frac{1}{\lambda} = \frac{2R}{v} e^{2G}$, $\ln(\tau) \sim 2G + \ln \frac{2R}{v}$)

trial frequency $f \sim \frac{v}{2R}$ & assumes indep. α . $\rightarrow$ tunneling



Volume term: When an assembly of nucleons of the same size is packed together into the smallest volume, each interior nucleon has a certain number of other nucleons in contact with it. So, this nuclear energy is proportional to the volume, i.e. $\propto R^3 \propto A$.

Surface term: Nucleons at the surface are more weakly bound than nucleons in the center. This term is negative and proportional to the surface area, $\propto R^2 \propto A^{2/3}$.

Coulomb term: The potential energy from each pair of protons. As this is a repulsive force, the binding energy is reduced. From the form of the Coulomb potential, $\propto \frac{Z^2}{R} \propto \frac{Z^2}{A^{1/3}}$.

Asymmetry term: Accounts for the Pauli exclusion principle. Unequal numbers of neutrons and protons imply filling higher energy levels for one type of particle, while leaving lower energy levels vacant for the other type. heavy nucl.: $n \neq p \rightarrow$ larger mean distance between $p \rightarrow$ less Coulomb.

Pairing term: Accounts for the tendency of proton pairs and neutron pairs to occur. An even number of particles is more stable than an odd number due to spin coupling

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even A : $p \uparrow \rightarrow p \uparrow$ & $n \uparrow \rightarrow n \uparrow$ & $p \downarrow \rightarrow p \downarrow$ & $n \downarrow \rightarrow n \downarrow$ or $p \downarrow \rightarrow p \downarrow$ & $n \downarrow$